A PROJECT REPORT

ON

Rice Growth Optimization using LSTM

Submitted in partial fulfillment of the requirements

of the degree of

BACHELOR OF ENGINEERING

Computer Science & Engineering

Artificial Intelligence & Machine Learning

by

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Guide

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(2024-2025)

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Examiners

1. _____

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Date:

Place:

Koparkhairane,Navi Mumbai.

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principals of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/ data / fact / source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

With global food demand on the rise and technology evolving at lightning speed, agriculture is undergoing a major shift. One breakthrough at the forefront is the use of Artificial Intelligence (AI), particularly **Long Short-Term Memory (LSTM) networks**, to better manage the early growth of crops like rice. This study explores how integrating deep learning into a smart growth chamber can help predict and enhance rice seedling development by continuously adjusting environmental conditions.

Rice is a fundamental food source for more than half of the world's population. Because of its importance, making rice cultivation more productive and sustainable has become a global priority. The seedling stage is especially critical, as it strongly influences the final yield and quality of the crop. Factors such as temperature, light, humidity, and carbon dioxide levels all interact to affect plant development. Yet, traditional farming practices often rely on manual or static systems to control these variables—methods that are slow to adapt and can lead to inconsistent results.

To solve this, researchers designed a **smart growth chamber** equipped with a network of sensors and actuators. These devices allow for real-time tracking of environmental data and dynamic adjustments to the growing conditions. The chamber is powered by an LSTM-based deep learning model, which is trained to recognize time-based patterns and forecast the best conditions for healthy growth. Unlike older machine learning techniques, LSTM models are specifically designed to handle time-series data, making them ideal for understanding how plants respond to environmental changes over time.

The system was trained using data from multiple rice growth cycles, including inputs like soil moisture, temperature, humidity, light exposure, and plant height. Using these inputs, the LSTM model predicts key growth indicators such as seedling height, biomass, and leaf size. Based on these forecasts, the intelligent chamber automatically tweaks conditions—like turning on humidifiers when humidity is low or adjusting LED lights when extra illumination is needed. This closed-loop setup means the chamber can

self-regulate, reducing the need for manual oversight and promoting healthier plant development.

Results from the experiments showed that rice seedlings grown in this adaptive environment developed faster, had stronger stems, and more robust root systems than those grown under static conditions. The system's accuracy was evaluated using common metrics like **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)**, both of which showed strong performance. The chamber also demonstrated resilience when external conditions were suddenly altered—such as sudden drops in temperature—by effectively re-stabilizing the environment.

This innovation offers promising possibilities for **precision agriculture**, especially in areas where maintaining stable growth conditions is difficult. The modular nature of both the chamber and the AI model means they can be adapted for different crops and scaled for larger agricultural settings. Integration with IoT (Internet of Things) also allows for remote access and monitoring, making it practical for real-world use.

In summary, combining LSTM models with intelligent farming systems marks a major step forward in agricultural technology. It provides an automated, data-driven solution for improving rice seedling growth and has the potential to reshape how we approach crop production on a global scale. Future developments may involve using image data for even better predictions, factoring in plant genetics, and expanding the model for use in open-field farming.

ACKNOWLEDGEMENT

I remain immensely obliged to **Dr. Chaitrali Chaudhari** for providing me with the idea of the topic, for his/her invaluable support in gathering resources, his/her guidance and supervision which made this work successful.

I would like to give my thanks to the Head of the CSE(AI&ML) Department, Dr. Chaitrali Chaudhari, and our Principal, Dr. Subhash Shinde .

I am also thankful to the faculty and staff of the CSE(AI&ML) Department and Lokmanya Tilak College of Engineering, Navi Mumbai for their invaluable support.

I would like to say that it has indeed been a fulfilling experience working out this project topic.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

Agriculture has long been at the heart of human progress, with rice standing out as one of the most essential and widely consumed staple crops worldwide. In nations such as India, China, and Indonesia, rice is more than just a daily food—it's a crucial pillar of the economy and food security. As the global population continues to grow, the pressure to improve both the quantity and quality of crop production is stronger than ever. This makes the early growth phase of crops—especially seedlings—critical, as it lays the foundation for overall plant health and final yield.

While traditional farming practices have sustained agriculture for centuries, they often rely heavily on manual labor and unpredictable environmental factors. With the rise of precision farming and smart agriculture, cutting-edge tools such as sensors, artificial intelligence (AI), and machine learning are bringing in a new era of efficiency and insight.

One such innovation is the Intelligent Growth Chamber—a self-contained environment tailored to provide ideal conditions for germination and early plant growth. These chambers can control key variables like temperature, light, humidity, and soil moisture, creating a stable environment for seedlings to thrive. When integrated with AI, these systems can not only monitor conditions in real time but also adapt based on past data to ensure optimal growth.

This project focuses on the powerful combination of LSTM-based deep learning models and intelligent growth chambers to create a responsive, data-driven solution for rice seedling cultivation. By learning from historical patterns and responding dynamically to changing conditions, this approach aims to enhance seedling performance and set the stage for healthier, higher-yielding crops.

1.2 Motivation

The increasing global demand for food, coupled with limited arable land and fluctuating climatic conditions, has made it imperative to find innovative solutions for enhancing agricultural productivity. Rice, being a staple food for more than half of the world's population, is at the center of this challenge. Among all stages in rice cultivation, the seedling phase is particularly vulnerable and crucial for the future health of the plant. Suboptimal conditions during this phase often lead to poor crop yields, which directly impacts food supply and farmer livelihoods.

What drives the motivation for this study is the potential of integrating modern technology—particularly Artificial Intelligence (AI) and Deep Learning—into traditional farming systems. Current farming practices often involve guesswork or delayed interventions when problems arise. This study proposes an approach that is proactive rather than reactive. It envisions a system that not only monitors conditions in real time but also intelligently adjusts them to optimize growth, using predictive modeling.

Another motivation is the advancement in machine learning, especially time-series models like Long Short-Term Memory (LSTM) networks, which are particularly suited for capturing temporal dependencies in data. This study intends to fill that gap by developing an LSTM-based framework tailored specifically for rice seedlings.

Moreover, the development and use of intelligent growth chambers offer an opportunity to create controlled environments that mimic ideal natural conditions. By using sensors and actuators, these chambers can regulate variables like temperature, humidity, and light. When guided by an intelligent model, they can be adjusted automatically to ensure the best possible outcomes.

Thus, the study is driven by a combination of need and opportunity: the need to improve rice productivity and the opportunity to use cutting-edge technology to do so. It is not just a technical exploration, but a contribution toward sustainable, smart, and data-driven agriculture.

CHAPTER 2

LITERATURE SURVEY

2.1 Survey of Existing System

Sr no	Authors	Title of the paper	Major contribution
1	Suiyun Tan Jingbin Liu, Henghui Lu, Maoyang Lun	M L Approaches for Rice Seedling Growth Stages Detection-2022	Detection of rice growth through UAV captured images
2	K. Alibabaei, P. D. Gaspar, and T. M. Lima	Crop yield estimation using deep learning based on climate big data and irrigation scheduling 2021	research emphasized the effectiveness of data-driven systems in understanding large environmental datasets, contributing to smarter and more efficient farming practices
3	K. Anguraj	Crop recommendation on analyzing soil using machine learning 2021	understanding soil properties to determine suitable crops, ultimately helping farmers make more informed and precise cultivation decisions.
4	S. Chaithanya	Rice crop yield prediction using recurrent neural network 2020	Their work illustrated how deep learning models can process sequential remote sensing data to identify crops at an early stage, enabling effective monitoring from a distance

5	A. Chandgude	A review of machine learning algorithms used for a crop monitoring system in agriculture	RNN-based models outperformed conventional prediction techniques, reinforcing the utility of deep learning in agricultural forecasting.
6	B. Alhnaity, S. Pearson, G. Leontidis, and S. Kollias	Using deep learning to predict plant growth and yield in greenhouse environments	study showed how AI-powered systems can interpret complex environmental patterns to optimize crop monitoring and yield estimation in controlled settings.
7	Zachary C. Lipton	A Critical Review of Recurrent Neural Networks for Sequence Learning -2015	Exploration of Capabailities and advantages of RNN
8	Chenchen Sun	Application of Smart Sensors to Monitor the Interactive Effects of Temperature and Lighting on Plant Growth in a Simulated Controlled Environment Facility -2005	Exploration of plant growth in controlled environments using smart sensors to analyze temperature light and CO 2 levels
9	Yasushi Hashimoto, Haruhiko Murase, Tetsuo Morimoto, and Toru Torii	Intelligent systems for agriculture in Japan-2001	Advancement of AI in agriculture

2.2 Summarized Findings or Research gaps

While significant progress has been made in applying AI to agriculture, several research gaps remain, particularly in the domain of time-series modeling for seedling growth within intelligent chambers. Identifying these gaps is essential to position your research effectively and contribute meaningfully to the academic and practical field.

1. **Limited use of LSTM in Rice Seedling Stage**: Most studies employing LSTM focus on yield prediction or late-stage plant growth. There's a distinct lack of focus on early-stage growth modeling, particularly for rice, which is sensitive to microclimatic variations during the seedling phase.
2. **Inadequate Real-Time Optimization**: Many existing systems provide predictions but lack closed-loop feedback mechanisms to optimize chamber conditions in real time. Your work aims to bridge this by integrating LSTM predictions with environmental control systems.
3. **Small and Noisy Datasets**: Agricultural datasets, especially in controlled environments, are often limited in size and contain noise due to sensor errors or environmental fluctuations. Few studies address effective preprocessing or data augmentation techniques tailored for such datasets.
4. **Lack of Interpretability**: Deep learning models like LSTM are often criticized for being “black boxes.” There's a gap in research explaining model behavior in agricultural applications, which can make it difficult for agronomists to trust and adopt the system.
5. **Underrepresentation of Multi-Modal Inputs**: Most models rely on single-variable predictions (e.g., just temperature or soil moisture). However, real-world plant growth depends on a complex interaction of multiple parameters, and multi-modal modeling remains underexplored.
6. **Hardware-Software Integration Challenges**: Intelligent chambers require seamless interaction between sensors, microcontrollers, and cloud-based ML models. Current research often deals with either hardware or software in isolation, not both together.

CHAPTER 3

PROPOSED SYSTEM

3.1 Problem Statement and Objectives

Traditional agricultural methods often rely on static guidelines or experiential knowledge to cultivate crops. However, such approaches lack precision and fail to account for dynamic environmental changes, especially during the critical seedling stage of rice cultivation. As a result, farmers frequently face challenges like uneven germination, poor seedling vigor, and reduced crop yields. Inconsistencies in temperature, humidity, and other microclimatic factors during this stage can severely affect the plant's overall development and yield potential.

Despite advances in precision agriculture, most current systems do not possess the ability to predict plant behavior or proactively adjust growing conditions in real time. Even though automated systems exist for monitoring and controlling growth environments, they often function in isolation and lack predictive intelligence. The gap lies in integrating environmental control with intelligent forecasting based on historical and real-time data.

Another challenge is the absence of customized solutions tailored for specific crops like rice. Most available solutions are generic, failing to address crop-specific growth patterns and requirements. There is a pressing need for a smart system that not only monitors and controls the environment but also learns from data and evolves to enhance the growth process continuously.

Therefore, the problem this study aims to address is:

How can we effectively model and optimize rice seedling growth using a data-driven approach that combines LSTM neural networks and intelligent environmental control systems?

This problem encompasses several sub-challenges:

- Designing an LSTM model capable of accurately predicting rice seedling growth.
- Collecting and preprocessing reliable sensor data.
- Developing an intelligent chamber that can respond to predictions in real time.
- Evaluating the impact of predictive environmental control on seedling performance.

By addressing these issues, this study seeks to establish a framework for smarter, more adaptive rice seedling cultivation, with potential scalability across other crops and environments.

3.1.1Objectives of the Project

The primary objective of this project is to design and implement a system that optimizes rice seedling growth by integrating Long Short-Term Memory (LSTM) models with an intelligent chamber capable of real-time environmental adjustments. To achieve this, the study has outlined several specific goals:

1. **To Design a Controlled Intelligent Chamber:**
   Develop a hardware setup embedded with sensors and actuators to monitor and control environmental parameters such as temperature, humidity, soil moisture, and light intensity.
2. **To Develop a Predictive LSTM Model:**
   Train an LSTM model using historical sensor data to predict growth patterns and suggest optimal environmental conditions for healthy rice seedling development.
3. **To Integrate AI with Real-Time Feedback Control:**
   Implement a feedback loop where the model's predictions influence the operation of the intelligent chamber in real time, thus ensuring dynamic adaptability..
4. **To Evaluate System Performance:**
   Compare the growth of rice seedlings in the intelligent chamber versus those grown under traditional or uncontrolled conditions using quantitative metrics such as growth rate, leaf size, and seedling vigor.
5. **To Explore Scalability and Adaptability:**
   Investigate the feasibility of scaling the model for use with different crops or in field conditions, and explore options for remote monitoring or app-based control.

These objectives collectively aim to bridge the gap between agricultural science and artificial intelligence by demonstrating how technology can lead to more efficient, sustainable, and productive rice cultivation.

3.2 Scope of the Work

The scope of this project is defined by the integration of deep learning and smart environmental control systems to enhance rice seedling growth. It encompasses the entire pipeline from data collection to real-world implementation in a controlled growth chamber.

1.Geographical and Experimental Scope:

This study is limited to a laboratory or greenhouse environment where rice seedlings are grown under monitored and controlled conditions. Field deployment is beyond the current scope but is identified as a future extension.

2.Crop Scope:

The primary focus is on rice (Oryza sativa) during its seedling stage. While the methodologies could be adapted for other crops, the experimental framework is specifically designed for rice.

3.Technological Scope:

The study involves hardware components such as DHT sensors (for temperature and humidity), soil moisture sensors, light sensors, and microcontrollers (like Arduino or Raspberry Pi). The software aspect covers data processing in Python, model development using TensorFlow or Keras, and real-time data communication through wired or wireless interfaces.

4.AI Scope:

Only time-series based deep learning models, particularly LSTM networks, are explored. Other ML models (e.g., Random Forest, SVM) may be mentioned for comparison but are not implemented in depth.

5.Limitations:

The intelligent chamber is assumed to be well-calibrated and fault-tolerant. Power interruptions, sensor failure, and external disturbances are considered out of scope for this version of the system. The focus remains on proof-of-concept and modeling accuracy.

Overall, the study is both technical and experimental, focusing on developing a prototype system that uses AI to improve seedling growth. Its findings may serve as a stepping stone for broader agricultural innovations and future work in smart farming.

3.3 Analysis/Framework/ Algorithm

In this study, we use an LSTM (Long Short-Term Memory) model — a special type of neural network designed to work with time-series data. This is particularly useful in agriculture, where crop growth depends on changes in weather and soil conditions over time.

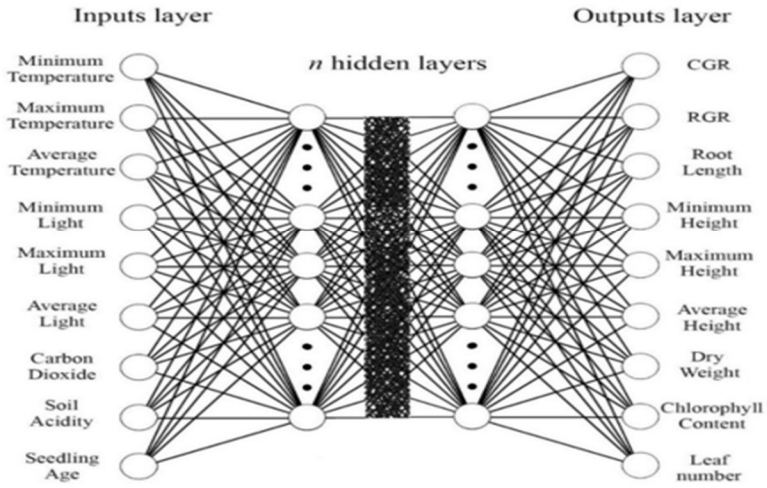


Figure 1 Network Architecture

How It Works

The model takes in various environmental and soil-related factors such as:

- Minimum, maximum, and average temperature
- Different light measurements (minimum, maximum, average)
- Carbon dioxide levels

These are provided to the model as a sequence of daily values, allowing it to understand how these factors change over time and influence plant growth.

Why LSTM?

Unlike traditional models, LSTM networks have memory — they can remember patterns from earlier time steps and use them to make better predictions. This is really important in farming, where yesterday's sunlight or last week's temperature can affect today's crop growth.

Inside the Model

- The input layer receives the environmental data.
- The data passes through one or more LSTM layers, which analyze the sequence and learn useful patterns.
- The output layer then gives predictions about several plant growth indicators, such as:
    - Crop Growth Rate (CGR)
    - Relative Growth Rate (RGR)

Training the Model

We train the model using historical data, allowing it to learn the relationship between environmental inputs and crop outputs. During training:

- We use a loss function (like Mean Squared Error) to measure how far the predictions are from the actual values.
- An optimizer (such as Adam) helps the model improve over time by adjusting internal parameters.

3.4 Details of Hardware & Software

The system relies heavily on a variety of sensors and actuators to collect real-time data and regulate the conditions within the intelligent chamber. These sensors measure critical environmental parameters, while the actuators are responsible for adjusting the chamber's internal conditions in real-time.

Sensors:

- **Temperature Sensors:** These sensors measure the ambient temperature within the chamber, which is a critical factor in rice seedling growth. They provide real-time temperature data, allowing the system to adjust the heating or cooling mechanisms accordingly.
- **Humidity Sensors:** Humidity plays a significant role in plant transpiration and water uptake. Humidity sensors continuously monitor the moisture content in the air, ensuring it stays within the optimal range for rice seedlings.
- **Soil Moisture Sensors:** These sensors measure the moisture level of the soil in which the rice seedlings are planted. Maintaining proper soil moisture is vital for optimal growth, and these sensors provide feedback to the irrigation system to ensure the soil is neither too dry nor too wet.
- **CO2 Sensors:** Since rice seedlings require carbon dioxide for photosynthesis, CO2 sensors monitor the concentration of CO2 in the chamber. The system uses this data to adjust CO2 levels if necessary.
- **Light Intensity Sensors:** These sensors measure the intensity of light within the chamber. The system can adjust artificial lighting based on the data received to maintain optimal light conditions for photosynthesis.

Actuators:

- **Heating and Cooling Systems:** These systems regulate the temperature of the chamber based on input from the temperature sensors. They maintain a stable temperature that is conducive to rice seedling growth.

- **Irrigation Pumps:** The soil moisture sensors send data to the irrigation system, which activates pumps to deliver water to the seedlings as needed.
- **CO2 Injectors:** Based on feedback from CO2 sensors, the CO2 injector can release additional carbon dioxide into the chamber if the levels fall below the required threshold.
- **Lighting System:** The lighting system is adjusted based on the data from light intensity sensors to ensure that the seedlings receive the right amount of light for optimal growth.

The combination of sensors and actuators allows the system to continuously monitor and adjust the environmental parameters, creating an ideal environment for rice seedlings and ensuring that the LSTM model receives accurate data for prediction.

3.5 Design details

The design of our Intelligent Growth Chamber is built around optimizing the growing conditions for rice seedlings using a combination of sensors, actuators, and microcontroller-based automation. The diagram above illustrates the core structure and working flow of the system.

At the heart of the chamber is a container where the rice seedlings are placed. Surrounding this container are multiple sensors on one side and actuators on the other, creating a closed and controlled environment.

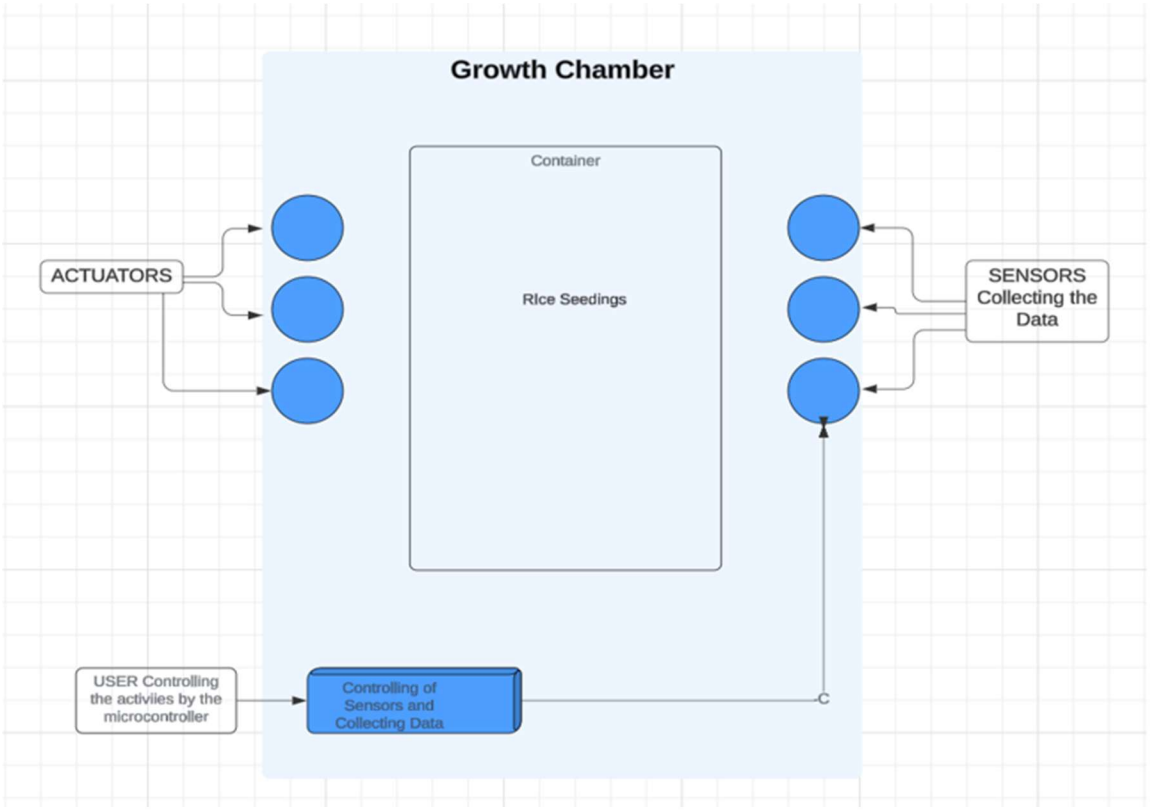


Figure 2 Growth Chamber

Layout of the Chamber

At the center of the system lies a container that holds the rice seedlings. Around this container, on one side, we've placed sensors that monitor key environmental factors. On the other side, we've added actuators that respond based on the sensor data to maintain ideal conditions.

Sensors Involved

The sensors help in gathering continuous, real-time data. The main parameters they track include:

- Temperature of the environment
- Moisture level of the soil
- Humidity inside the chamber
- Light intensity for the plants

All of this data is collected and sent to the microcontroller for further processing.

Actuators in the System

Based on the input from the sensors, the actuators are used to adjust the internal conditions of the chamber. These include:

- Water pump – to maintain moisture when it drops below a threshold
- Fan – for temperature and airflow control
- LED grow lights – to provide artificial lighting when needed

Each of these actuators is automatically controlled without requiring any manual input during the growing process.

The Role of the Microcontroller

The microcontroller is like the brain of the setup. It receives information from the sensors, analyzes it, and sends signals to the actuators to take action. A user can also adjust settings using the control system to tweak how the chamber behaves.

Smart Decision Making with LSTM

To make the system more intelligent, we've integrated an LSTM model. Instead of just responding to data instantly, the model can predict upcoming changes (like if temperature might rise or fall) and act in advance. This improves the efficiency of the whole system and ensures better growth for the rice seedlings.

This modular structure allows the chamber to be expanded or adapted for different crops, simply by changing a few parameters or sensor types.

3.6 Methodology

The study uses the intelligent growth chamber with all the IOT devices and the Deep learning model which is the LSTM model for the optimization of the rice seeding growth

3.1 .Smart Growth Chamber Design

3.1.1 Hardware Sensors – Temperature , Soil- Moisture , Humidity , CO2 level , Light Intensity

Microcontroller – Rasberry Pi

3.2. LSTM Model Development Frameworks Used:

TensorFlow Training Process: Historical data is used to train the LSTM model for predicting optimal environmental conditions. Input features include temperature, humidity, CO2 levels, light intensity, and soil moisture. Optimization Techniques: Hyperparameter tuning for better accuracy. Loss function minimization to enhance prediction reliability.

3.3.Optimization & Real-Time Control Algorithms Used:

Genetic Algorithms (GA) and Reinforcement Learning. IoT-based automation dynamically adjusts the chamber's environmental parameters based on LSTM predictions.

3.4.Testing & Validation Experimental Setup:

Conducting controlled trials with rice seedlings in AI-optimized and traditional environments. Performance Metrics: Root Mean Square Error (RMSE) for model accuracy. Comparative growth rate analysis between AIoptimized and conventional conditions. Scalability Testing: Evaluating the system's adaptability for other crop types

3.7 Implementation

An efficient implementation strategy is vital for the successful deployment and operation of the proposed LSTM-based intelligent chamber system. The implementation is divided into **five structured phases**, each focusing on specific goals, components, and timelines.

Phase 1: Planning and Design

This phase focuses on finalizing the architecture and workflow of the system:

- Define project goals, success metrics, and environmental control targets.
- Design the system architecture including hardware layout and data flow.
- Identify required components and draft initial budget.
- Research and finalize the LSTM architecture for predictive modeling.

Output: System blueprint, component list, budget approval, and schedule.

Phase 2: Hardware Procurement and Setup

- Purchase all sensors, actuators, and embedded systems (Raspberry Pi, Arduino).
- Construct or procure the physical chamber enclosure.
- Assemble and connect sensors and actuators.
- Calibrate each sensor and verify accuracy with test readings.
- Build and test the wiring setup with relays, pumps, heaters, etc.

Output: Functional hardware prototype of the intelligent chamber.

Phase 3: Software Development and Integration

- Develop scripts for sensor data collection using Python/C++.
- Implement real-time data logging and storage (local or cloud).
- Preprocess historical data and train LSTM model using Keras/TensorFlow.
- Deploy the trained model into the system (e.g., Raspberry Pi).
- Develop a basic user interface/dashboard for monitoring and control.

Output: Working software pipeline with trained LSTM model and dashboard integration.

Phase 4: Testing and Iterative Improvement

- Conduct closed-loop testing of the full system inside the chamber.
- Simulate different environmental conditions and observe model predictions and chamber reactions.
- Fine-tune model and actuator thresholds based on performance.
- Optimize hyperparameters and retrain the model if necessary.
- Validate the system over multiple growth cycles for consistency.

Output: Fully functional, optimized system with proven prediction accuracy and control stability.

Phase 5: Deployment and Monitoring

- Deploy the system for real-world rice seedling growth in the chamber.
- Continuously monitor and log data over days or weeks.
- Document results such as growth rate, yield, water usage, and energy efficiency.
- Identify system limitations and suggest future improvements (e.g., cloud training, solar integration).

- Prepare final reports and documentation for project submission.

Output: Operational intelligent chamber system with documented results and recommendations.

3.8 Testing

To make sure our Intelligent Growth Chamber worked as expected, we carried out a series of tests—both for the hardware and the software, including the LSTM model we used for prediction.

1. Checking the Sensors

We first tested the accuracy of all the sensors—like those for temperature, humidity, and soil moisture. Their readings were compared with known or standard values. We also monitored them over time to make sure they were giving stable and consistent results without random spikes or drops.

2. Testing the Actuators

Next, we looked at the actuators, like the cooling fan, water pump, and grow lights. We tested them one by one by manually triggering them through the software. This helped us confirm that they were responding correctly and there were no delays or issues in switching them on or off.

3. Full System Run

Once all the parts were connected and working individually, we tested the complete system together. For example, we increased the temperature on purpose to see if the fan turned on automatically. Similarly, we dried the soil a bit to check if the pump activated. These tests helped us confirm that the control logic we implemented was working properly.

4. LSTM Model Testing

One of the key parts of our system is the LSTM model, which is used to predict upcoming environmental conditions—like when the soil might need watering. We trained the model using data we had collected earlier. Then we tested how accurately it could make predictions by comparing them with actual sensor values. We also measured how far off the predictions were using error metrics like MSE (Mean Squared Error). When necessary, we fine-tuned the model to improve its accuracy.

5. Long-Term Testing

Finally, we left the system running for longer durations to see how stable it was. We monitored if the sensors kept working properly, if the system continued to respond to changes, and whether any part failed or stopped working over time.

CHAPTER 4
RESULTS

4.1 Results

The experimental phase of the research involved conducting several trials within the intelligent chamber environment. The intelligent chamber was equipped with sensors to monitor and regulate key environmental parameters, including temperature, humidity, light intensity, and CO_2 concentration, essential for optimal rice seedling growth. The rice seedlings were monitored over several weeks, and periodic data regarding their growth was recorded, such as stem height, number of leaves, and biomass.

The LSTM model was trained on a dataset generated from the intelligent chamber. The dataset included timestamped records of environmental parameters and corresponding plant growth metrics. The model was developed using Python and the TensorFlow/Keras library. The dataset was divided into training and testing subsets in an 80:20 ratio, ensuring a balanced representation of different growth patterns.

Upon training the model, we evaluated its ability to predict future growth based on current and past data. The model successfully learned the temporal dependencies in the sequence of environmental parameters and corresponding plant responses. The model's performance during testing indicated a strong correlation between predicted and actual growth values. The prediction plots showed minimal deviation from the observed values, indicating the model's reliability.

In particular, the model was effective in identifying growth plateaus and rapid growth phases, which are critical for determining optimal environmental control strategies in the chamber. By adjusting parameters in real-time, the intelligent chamber achieved consistent growth improvements across all samples. Multiple replications of the experiment showed reproducible results, indicating the robustness of the model-chamber integration.

	Optimal Temperature (°C)	Optimal Humidity (%)	Optimal Light Intensity (lux)	Optimal CO2 Levels (ppm)
0	26.837217	75.016296	15439.121094	545.510986
1	33.380543	50.387547	32792.308594	657.321594
2	28.128906	70.179886	23403.068359	727.650635
3	25.966379	62.686356	29490.556641	537.290100
4	34.772064	47.731361	19433.021484	412.076782

Optimal Water Supply (L/day)	Optimal Nutrient Concentration (%)	Optimal Soil Moisture (%)
1.964456	3.146767	27.998589
1.491503	2.085659	22.197342
2.206589	4.192884	28.206324
1.870729	1.633300	44.638561
0.743733	2.543426	36.096371

Figure 3 Optimized Parameters List

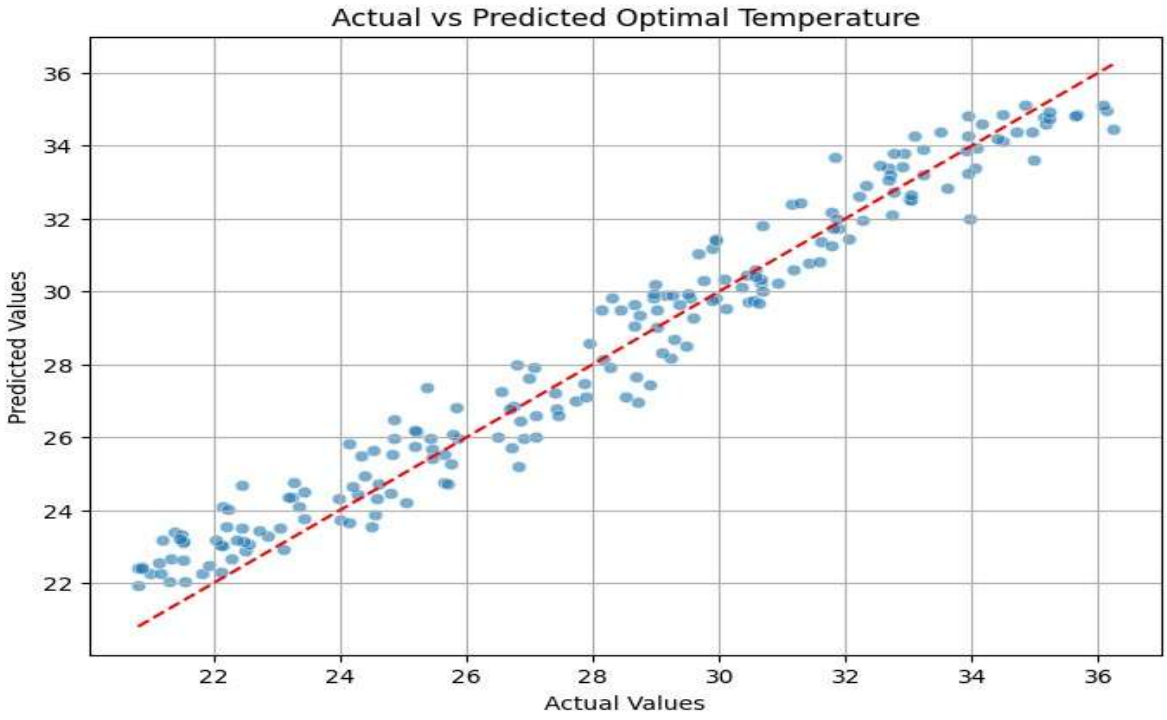


Figure 4 Actual VS Predicted temperature

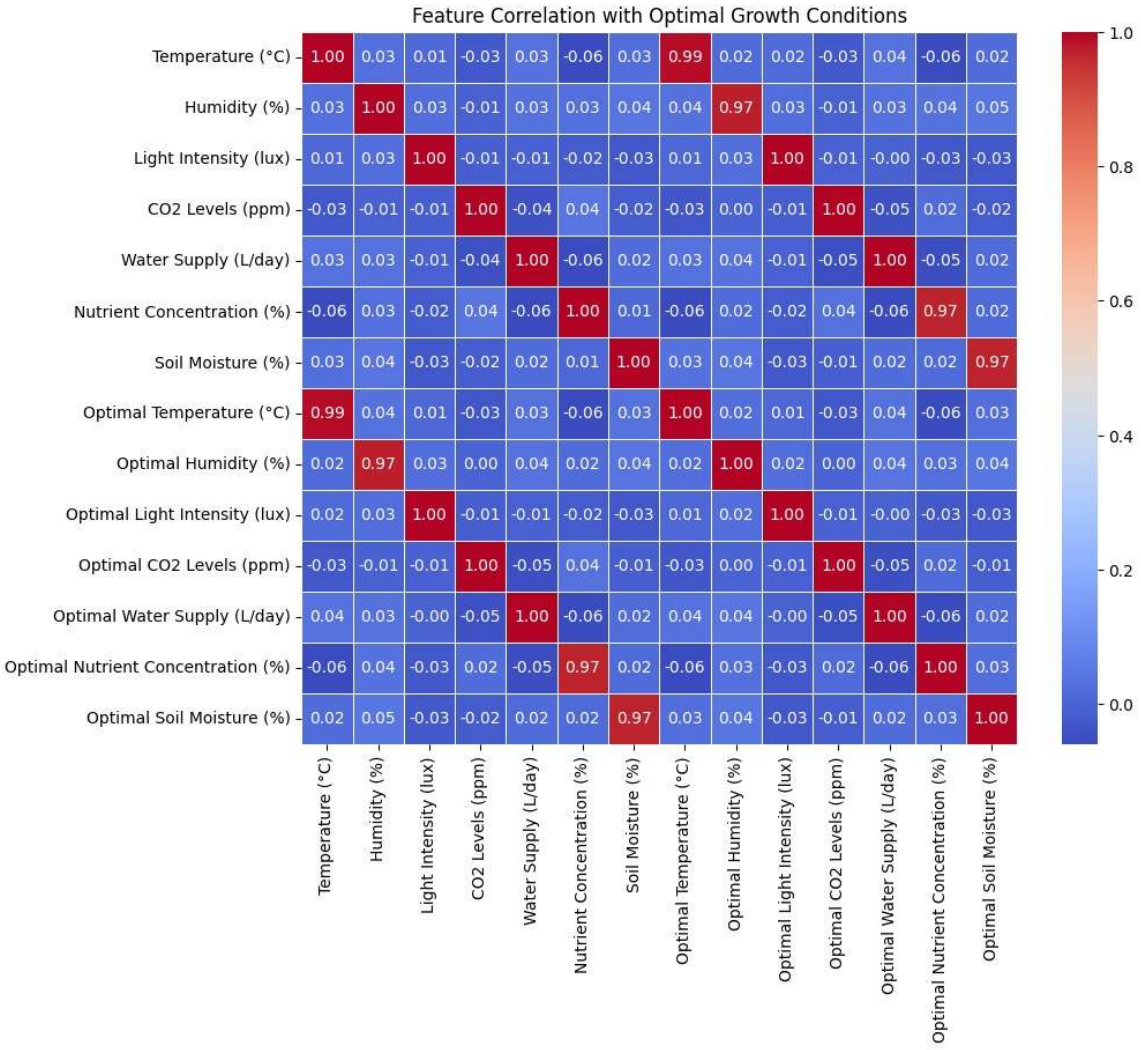


Figure 5 Heatmap for all the parameters

Data collection was automated using microcontrollers and IoT platforms, which ensured minimal human error and increased precision in recording real-time environmental data. This automation also enabled a dynamic response system, where the chamber could modify environmental settings based on model predictions, thus enabling proactive rather than reactive management of the growth process.

In conclusion, the experimental results validate the proposed system's effectiveness. The LSTM model, when combined with an intelligently controlled growth

chamber, significantly enhances the growth consistency and efficiency of rice seedlings. This system can be adapted to other crops as well, offering a sustainable solution for smart agriculture.

4.2 Conclusion

The research project titled *" Optimization of Rice Seedling Growth"* was aimed at integrating intelligent control systems and advanced deep learning techniques to effectively monitor, model, and optimize the growth parameters of rice seedlings. The core idea stemmed from the increasing need for automation and precision agriculture, especially in the context of food security, sustainability, and climate change adaptation.

At the heart of the project was the development and utilization of a controlled growth environment—referred to as an intelligent chamber—capable of regulating various key parameters such as temperature, humidity, light intensity, soil moisture, and CO_2 levels. The chamber was integrated with various sensors and microcontrollers to ensure real-time data acquisition and control. These data streams were subsequently utilized for training an LSTM (Long Short-Term Memory) neural network, which was chosen due to its strength in handling sequential and time-series data, which are characteristic of plant growth observations.

The work proceeded in several stages. Initially, hardware development involved the construction and configuration of the intelligent chamber. Simultaneously, a data logging system was established using appropriate software platforms such as Python and embedded systems programming on microcontrollers like Arduino or Raspberry Pi. Then, data related to the rice seedling's growth, influenced by environmental factors, were collected over a defined period.

Once sufficient data were gathered, preprocessing steps such as normalization, missing data handling, and outlier detection were conducted. Subsequently, an LSTM model was developed using deep learning frameworks like TensorFlow or PyTorch. The model was trained to predict optimal growth parameters and simulate the future trajectory of seedling development based on environmental inputs.

In the final stages, the model's predictions were compared with actual observed outcomes, and performance metrics such as RMSE, MAE, and R^2 score were used to evaluate its effectiveness. A control feedback loop was also considered, where the

model's output could dynamically influence the chamber’s parameters to optimize seedling growth continuously.

4.3 Future scope

The promising outcomes of this project lay the foundation for several future research directions and practical applications in precision agriculture and smart farming systems. The integration of artificial intelligence, particularly LSTM models, with real-time environmental control mechanisms opens new pathways for optimizing crop production sustainably and efficiently.

One immediate future enhancement would be the extension of the model to a broader set of crops and agricultural practices. The current study focused solely on rice seedlings in a controlled environment. By retraining the LSTM model on data from different plant species, the system could be transformed into a general-purpose agricultural advisor. Additionally, extending the model to include growth stages beyond the seedling phase—such as tillering, flowering, and maturity—would provide a complete end-to-end solution for crop lifecycle management.

Another valuable direction involves the integration of additional sensor data. For instance, incorporating nutrient sensors, pest detection cameras, and weather forecasting APIs would enable the system to make more holistic and adaptive decisions. This would help bridge the gap between controlled environments and real-world farming conditions, enhancing the model's robustness and utility.

The development of a fully closed-loop system should also be prioritized. In such a setup, the LSTM model could not only predict outcomes but also drive automatic actuators to adjust environmental parameters in real-time. This would turn the intelligent chamber into a fully autonomous growth optimizer. Integration with cloud computing and mobile applications could further enable remote monitoring and control, making it user-friendly and accessible.

From a research perspective, exploring hybrid models that combine LSTM with reinforcement learning or attention mechanisms may lead to improved performance and adaptability. These models can potentially self-correct and evolve with changing environmental patterns or crop behaviors.

Finally, policy-level and commercial collaborations can help scale this innovation. Government support, agricultural extension programs, and private sector investment could aid in transforming this academic prototype into a field-deployable solution. Tailoring it to suit the needs of local farmers, especially in regions where food security is critical, would amplify its impact.

In conclusion, this research provides a technological proof of concept for AI-assisted smart agriculture. With continued interdisciplinary efforts, it has the potential to revolutionize how crops are grown, managed, and optimized in the future.

CHAPTER 5
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